

10-11

$$d = 0.025 \text{ m} \quad l = 0.2 \text{ m} \quad t_w = 100^\circ \text{C} \quad t_\infty = 20^\circ \text{C} \quad u_\infty = 5 \text{ m/s}$$

迭代算法 $t_f'' = 40^\circ \text{C}$

$$t_f = \frac{1}{2}(t_f' + t_f'') = \frac{1}{2}(20 + 40) = 30^\circ \text{C}$$

$$t_f = 30^\circ \text{C}, \quad \rho = 1.165 \text{ kg/m}^3, \quad c_p = 1.005 \text{ kJ/kg}\cdot\text{K} \quad \lambda = 2.83 \times 10^{-2} \text{ W/m}\cdot\text{K}$$

$$\mu = 18.6 \times 10^{-6} \text{ kg/m}\cdot\text{s} \quad \nu = 16 \times 10^{-6} \text{ m}^2/\text{s} \quad Pr = 0.701 < 10 \Rightarrow m = 0.37$$

$$Pr_w = 0.688 \text{ (100}^\circ\text{C)}$$

$$\text{A. A. Zhukauskas} \quad Nu = C Re^n Pr^m (Pr/Pr_w)^{1/4}$$

$$Re = \frac{u_\infty d}{\nu} = \frac{5(0.025)}{16 \times 10^{-6}} = 7812.5 \quad 10^3 < Re < 2 \times 10^5 \Rightarrow C = 0.26, n = 0.6$$

$$Nu = 0.26 (7812.5)^{0.6} (0.701)^{0.37} (0.701/0.688)^{0.25} = 48.613$$

$$h = Nu \frac{\lambda}{d} = 56.168 \text{ W/m}^2\cdot\text{K}$$

$$\Phi = \pi d h l \Delta t = 61.3308 \text{ W}$$

$$\text{等壁温} \quad \Delta t = \frac{\Delta t' - \Delta t''}{\ln \frac{\Delta t'}{\Delta t''}} = 67.5212^\circ \text{C}$$

$$\Delta t' = t_w - t_f' = 80^\circ \text{C}$$

$$\Delta t'' = t_w - t_f'' = 60^\circ \text{C}$$

$$\Phi_r = \rho u_\infty A c_p (t_f'' - t_f') = 57.4727 \text{ W} \approx \Phi \Rightarrow t_f'' = 40^\circ \text{C}$$

10-13

$$d = 0.076 \text{ m} \quad t_w = 80^\circ \text{C} \quad t_\infty = 20^\circ \text{C}$$

$$t_m = \frac{t_w + t_\infty}{2} = 50^\circ \text{C}$$

$$\rho = 1.013 \text{ kg/m}^3 \quad c_p = 1005 \text{ J/kg}\cdot\text{K} \quad \lambda = 2.83 \times 10^{-2} \text{ W/m}\cdot\text{K}$$

$$\nu = 17.85 \times 10^{-6} \text{ m}^2/\text{s} \quad Pr_f = 0.698$$

$$\alpha = \frac{1}{T_m} = \frac{1}{273.15 + 50} = 0.003075 \text{ K}^{-1}$$

$$g = 9.81 \text{ m/s}^2 \quad \Delta t = t_w - t_\infty = 60^\circ \text{C}$$

$$Nu_f = C(GrPr)^n \quad Gr = \frac{\alpha g \Delta t d^3}{\nu^2}, 2481948.594$$

$$GrPr = 1732400 \cdot 118 = 1.7324 \times 10^6 \quad 10^4 < GrPr < 5.76 \times 10^8$$

$$\Rightarrow C = 0.48 \quad n = 0.25$$

$$Nu_f = 0.48(1.7324 \times 10^6)^{0.25} = 17.4142$$

$$h = Nu_f \frac{\lambda}{d} = 6.4845 \text{ W/m}^2 \cdot \text{K}$$

$$\Phi = \pi d h \Delta t = 92.8948 \text{ W/m}$$

10-14

$$d = 0.15 \text{ m}$$

$$H = 1.6 \text{ m}$$

$$L = 5 \text{ m}$$

$$t_{w1} = 150^\circ \text{C}$$

$$t_{w2} = 100^\circ \text{C}$$

$$t_{\infty} = 18^\circ \text{C}$$

$$\text{垂直段: } t_{m,t} = \frac{t_{w1} + t_{w2}}{2} = 84^\circ \text{C}$$

$$\lambda_v = 3.08 \times 10^{-2} \text{ W/m} \cdot \text{K}$$

$$\nu_v = 21.494 \times 10^{-6} \text{ m}^2/\text{s}$$

$$g = 9.81 \text{ m/s}^2$$

$$\alpha_v = \frac{1}{t_m} = 0.002363 \text{ K}^{-1}$$

$$Pr_v = 0.612 \quad \Delta t_w = 132^\circ \text{C}$$

$$Gr_v = \frac{\alpha g \Delta t_v H^3}{\nu_v^2} = 2.7128896 \times 10^{10}$$

$$Gr_v Pr_v = 1.875149285 \times 10^{10}$$

$$Gr_v Pr_v > 4.65 \times 10^9 \quad C = 0.1 \quad n = 1/3$$

$$Nu_v = C(GrPr)^n = 0.1(1.875149285 \times 10^{10})^{1/3} = 265.6717$$

$$h_v = Nu_v \frac{\lambda_v}{H} = 5.1175 \text{ W/m}^2 \cdot \text{K}$$

$$\Phi_v = h_v \pi d H \Delta t_v (3600) = 1833.5611 \text{ kJ}$$

$$\text{水平段: } t_{m,H} = 59^\circ \text{C} \quad \alpha = \frac{1}{t_{m,H}} = 0.0030107 \text{ K}^{-1} \quad \Delta t_H = 82^\circ \text{C}$$

$$\lambda_H = 2.893 \times 10^{-2} \text{ W/m} \cdot \text{K}$$

$$\nu_H = 18.868 \times 10^{-6} \text{ m}^2/\text{s}$$

$$Pr_H = 0.6162$$

$$Gr_H = 2.296002647 \times 10^7$$

$$Gr_H Pr_H = 1.598477043 \times 10^7$$

$$10^7 < Gr_H Pr_H < 10^{12} \Rightarrow C = 0.125 \quad n = 1/3$$

$$Nu_{f,H} = 0.125 (Gr_H Pr_H)^{1/3} = 31.48803$$

$$h_H = 6.073 \text{ W/m}^2\cdot\text{K}$$

$$\dot{Q}_H = \pi d h_H \Delta t_H (360^\circ) = 4224.067 \text{ kJ}$$

$$|\dot{Q}_{tot}| = 6057.678 \text{ kJ}$$